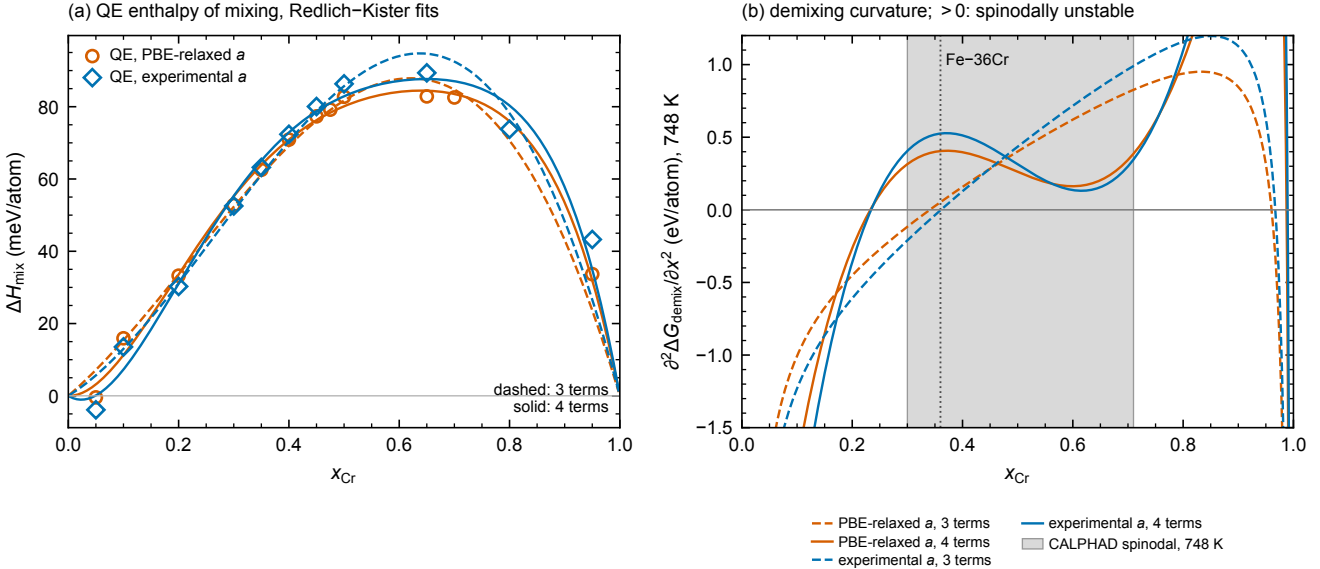
(a) QE enthalpy of mixing, Redlich−Kister fits
QE, PBE-relaxed a
QE, experimental a
ΔHmix (meV/atom)
xCr
dashed: 3 terms
solid: 4 terms
(b) demixing curvature; > 0: spinodally unstable
Fe−36Cr
∂²ΔGdemix/∂x² (eV/atom), 748 K
xCr
PBE-relaxed a, 3 terms
PBE-relaxed a, 4 terms
experimental a, 3 terms
experimental a, 4 terms
CALPHAD spinodal, 748 K